

ActInf Textbook Group ~ Cohort 2 ~ Meeting 8

(Chapter 4, part 1)

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greetings everyone it is uh October 21st 2022. we're in textbook group number 50.1 no I just have a slight typo in my OBS it's cohort 2 and meeting eight we're discussing chapter four in our first discussion of chapter four we're going to as we did not have any chapter or for specific questions from cohort two today largely we're going to be reviewing and revising questions that were initially addressed in cohort one taking a second pass and seeing where we can like add some context and simple or complex or however questions that we can add so you can be adding in the chat or in the questions tables directly or like just unmuting and um addressing them so that we can have high quality and quantity of a large diversity of questions on the material that people are curious about when they're reviewing the text before we go to the questions table does anyone have any general thoughts or comments on chapter four yes please Ollie as we discussed in the previous cohort this chapter is probably the most technical one in the whole book so uh as people go through the formalism here in this chapter uh I think it's quite normal to not fully understand every single equation there so uh in case uh there are some difficulties in getting uh in getting the content of this chapter I think some of these difficulties not all of them would be unpacked in the later chapters thank you very much totally agreed anyone else want to make a general chapter 4 comment yeah there's such an interesting rhythm chapter one being largely contextualizing and framing the chapter 2 3 dialectic the low road and the high road chapter four with a focus on the generative model and it is a very technical chapter five whiplashing over to neurobiology and less on the fundamental formalisms more on the neuroviological relationships that constitutes the epistemic first half of the book the second half of the book has kind of a rhyming pattern but less focused on epistemic and more on pragmatic although the pragma of epistem whereas chapter 6 is focusing more on the recipe for making the model and less of the contextualizing then there's another chapter pair seven and eight discrete and continuous time and then chapter nine which is where we're heading into in just um 50 minutes in the first cohort with model

based data analysis and then chapter 10 as sort of like the symmetry of chapter one and the three appendices okay if anyone has comments of course just like raise your hand or speak up or put it in the chat or however all on upvote these so we can upload them when we've gone through them today well let's scroll through the chapter before we go to the questions just to get a view of it this chapter complements the preceding chapters conceptual treatment of active with a more formal treatment it sets out the relationship between free energy Bayesian inference form of generative models and the Dynamics obtained from minimizing free energy the key focus is on how time is represented 4.2 Bayesian inference to free energy equation four one recalls Bayes theorem figure four one describes Jensen inequality which is related to the tonicity of the natural log or just the logarithm function in general and it's uh pretty fundamental point and it speaks to the advantage of using for example log probabilities and the different kinds of operations that are possible on logged values here is an application of Jensen's to derive a free energy value as a bounded um is it bounding foreign ments amongst Bayes theorem and free energy generative models to calculate the free energy we need three things data a family of variational distributions and a generative Model comprising A prior and likelihood the section has two types of generative model the first is dealing with categorical variables the second deals with continuous variables bigger 4.2 is helping gain some familiarity with like nodes and edges in a Bayesian graph so here in a base graph nodes are variables circular nodes are variables and edges which are like labeled with the with a box reflect their relationships so one can imagine and we'll return to this in the questions these map onto situations where like there's a hidden cause and an observation you know temperature in the room thermometer reading two things that influence one outcome one hidden thing that influences two outcomes a hierarchical model where something changes something that results in an outcome and so on and then the boxes are labeled according to the way that we might see that in a more uh formulaic conditional likelihood expression vertical line means conditional upon so like box too is the probability distribution of Y conditioned upon X because of the directedness of this error figure 4 3 is a very central figure in the textbook and it sets up the similarities and differences largely between the discrete time and the continuous time formulations of active inference generative models in terms of partially observable Markov decision processes through discrete time and for the um more uh Taylor series expansion-like approach used for continuous time does anyone want to like give a thought or an overview on what is happening in figure four three yes Ali and then anyone else well the top figure is um what happens when we uh uh when we want to model uh discrete time uh discrete Time Event

and the bottom one is more conducive to modeling The Continuous time phenomena so as we see you see they're quite parallel and the only difference or at least the most important difference here is the role that policies play in the discrete time case and the V play and the continuous time case and the policy is denoted with the π and the top figure and it relates to the top figure relates to the Palm DP approach to modeling the discrete time phenomena as we'll see at length in chapter seven and the bottom one is unpacked in more I mean in chapter 8 I guess excellent again anyone can uh raise their hand and go for it but just to kind of describe what the nodes and edges are here we'll be thinking about like the temperature in the room is the hidden State and the thermometer is the observation so in the discrete time formulation this is the temperature in the room through time it's the hidden State it's in the generative process then there's an a matrix here labeled as 2 which is the distribution of observations on the thermometer conditioned upon the temperature at that time so the a matrix can be used as a recognition density to do inference on the hidden State given the observation or it can be used in a generative capacity given a temperature to emit distributions of thermometer observations the 3 corresponds to the B Matrix which is the distribution of the state at the next time conditioned upon the state at the given time and the policy π so 3 or B describes how the temperature is changing Through Time so this kind of through line of hidden State and then a b Matrix or three and then a hidden State this is something changing through time and then π is policy which is uh a function of the decision making of the entity and it influences how States change their time that's the discrete time formulation time Point negative one zero and one or however and often you see a d or a prior which is uh how the first hidden State value gets initialized that's the discrete time formulation in the continuous time formulation they're being laid out side by side because the isomorphy of the models is apparent they have the same topological structure however there are some really important differences rather than describing how a hidden State changes through time at explicitly demarcated time points like minute 12 minute 13 minute 14. um what's Happening Here is that the the value of the Hidden State at the focal time is still in an a matrix relationship with observables so the probability of Y given X is exactly analogous to this one of the observations conditioned on the hidden state however taking the place of the B Matrix three above here which describes how hidden States change their time in the discrete time formulation is actually a edge that describes the derivative of x conditioned upon X and V which relates to slowly changing causes in the world analogous to policy and so this is actually not doing explicit prediction of future explicit time points but rather it's doing

something like a Taylor series expansion where as you take more and more derivatives further and further away from a focal point you gain better and better approximation through Power series expansion using higher and higher derivatives so here's like an example here's a sine wave and the first derivative just gives you this locally linearizable range but there's a well-defined computational complexity for a Taylor series you can say the zeroth order is just zero in the first order just a linear regression and then there's just a quadratic and then there's just a third power relationship and so on and so as you get more and more getting into the blue and the purple and so on you get better and better fit and you extend the range of fit further and further out and it always gets strictly better this is going to come back in chapter seven and eight which are going to focus more directly on the discrete and continuous time formulations but for here and four they're just being introduced as different ways that we can have a generative model in active so now figure four or section four four is gonna come to the top of four three there's some formalisms related to categorical distributions that are compatible with partially observable Markov decision processes here's an expected free energy G that describes some different partitionings on as a function of policies evaluating policies based upon their impact on the pomdp four key ingredients for the discrete time generative model a b c d then moving to continuous time but first there is a kind of a technical interlude on Markov blankets previously there was a box defining the markout blanket and here it's discussed in the context of message passing yes GSF anywhere if you want to call attention to yeah [Music] um I was going back to the figure four three so in looking at the at the parallel between the top and the bottom figure so in the in the top figure π that is the policy is obviously something that is selected by the agent right what about new in the like I mean the the Greek letter that that is uh that is paralleled to π in the bottom figure uh new is it um because he said slowly slowly varying causes in in the environment or something like that but is that something emitted or selected by the agent or is it something because does anyone have a thought on this the distinction is just between discrete and continuous is not between like something that that it's a autonomous autonomously selected by the agent or something that is you know happening in the environment right huh Ali uh actually in continuous um in continuous time situation uh the new here can also be related to uh the non-cognitive aspects of the system such as the attractors of the system so that's why it's described in terms of the slowly varying causes for instance in chapter 8 we will see that uh well we have some static variables According to which the system behaves so that's not exactly something that the system will choose per se it's some some some variables or some states towards which the system that will approach so

that's yeah that's why uh they've termed it in battery so for instance when applying to to movements that can be some sort of anatomical constraint of the movement something like that exactly yeah I see okay thank you thank you for um sharing that Ali it reminds me of the question that came later action is part of in this is chapter eight action is part of the generative process not the generative model and so the actual interpretation of slowly varying causes and the mapping of slowly varying causes to real world phenomena is one interesting area another General point is that uh if we look to the figures we'll see that the first figure is the action perception Loop and initially there's some usage of the base graph in terms of elaboration on the action perception Loop and so on but when we get to chapter four and the we don't see the action perception Loop represented the same way and so it's an interesting question to ask where these variables map onto for example The Entity model this isn't simply an entity model okay so it's completing our our quick scan before we go to the questions message passing and Mark all blankets this is definitely an area that we could learn a lot more about and this is also an area where software packages exist with the work of the bias lab and therefore knee package figure four four Bayesian message passing does anyone want to describe this figure Just one thought well the overall thought is is this supposed to be something that we're supposed to in detail understand or is it more like an illustrative representation I was thinking exactly the same when I saw it it's like here's the kind of food you could make or is this like here's how you make this one dish um there are uh also it's not red from left to right per se at the level of the figure the caption begins with a right so starting with the left the scheme could be expanded hierarchically collapsing over time steps and policies for Simplicity so just looking at the left one here the subscripts have I on the bottom then the subscripts have I plus one but you see the same squiggle Π s ϵ so this is showing that within the context of a hierarchical model you can have recurring patterns of topological relationships of a base graph to enable for example hierarchical predictive processing active Bayesian action selection Etc models and then that is collapsed over time steps and policies so what's happening here so this is going to be unpacking within one layer on the right side and we see like so here there's no superscript but we can just take that to be like Π I so this is going to be this quartet being expanded out so here's o tilde like o Through Time coming in and here's like the actual discrete time observations coming in here's ϵ and S in a um uh predictive processing capacity and these are illustrating with labeled edges some of them the relationships between these different variables squiggle in this case that's funny when you search for this wiggle it finds this the it finds the other Sigma as well squiggle sub Π t is not two natural logs being

subtracted a natural log subtracting another natural log so what happens when you subtract what's $\log a$ minus $\log B$ it's the same as $\log a$ divided by B $\log$ open parenthesis a divided by B so this is Computing a ratio between the observation and the preference if that is one then the incoming observations are exactly the preferences the preference distribution if there's a surplus then the observation is higher if there's a deficit below one then it means that it's lower I hope that is correct way to say it so here's message passing and relationships amongst variables and we see that kind of like with an Epsilon intermediating we see the S relating to the O at a given time so this is a lot like the a matrix and here's the role of policy now we're in the um continuous time 4.5 Ali please I think this figure can be usefully compared with figure 7.12 because uh I believe uh the figure we'll see in chapter 7 7.12 I guess yes that's it is the uh I mean hierarchy the other interpretation of this uh figure from chapter four uh in terms of its uh hierarchical relationships but uh from graphical probabilistic modeling point of view and the other thing is uh that uh I mean the final position Sigma uh I don't think it's defined explicitly in the uh text but it's been defined as the expected prediction error in the actor in The step-by-step tutorial paper so uh yes that's also I wanted to mention that definition as well thank you this step-by-step tutorial which is also a four-part model stream is very very useful in some ways it is almost textbook length whereas in chapter four there was first the general discussion of Bayes graphs like again one cause one observation one cause two observations two to one one to one to and then instantly into the juxtaposition between discrete and continuous time as we've looked through today in contrast the approach taken in step by step first off is to only focus on the discrete time formulation but then they walk through a sequence of graphs and connect them to like increasing levels of complexity building towards the one that we see in 4.3 it's a little later than one might expect in the paper sorry about that here it is starting with just a simple graph that is exactly analogous to the one on the top left so they start with a static perception in the step by step they introduced this kind of e motif by showing a prior and three time points two changes observations mediated by the a this is corresponding to the music listening example so it's going to come up later in the textbook then policy as a way to influence state transition matrixes is introduced on the top right and free energy as an objective function Guided by preferences to do policy selection as inference and then finally on the bottom right there is some Precision on policies which is just a bringing in another variable beta and gamma which have like a inverse relationship to each other so that you can talk about like High precision as low ambiguity and vice versa so it's different ordering of talking about generative models okay we're in the continuous time branch I'm just

copying a link that Ali sent because I I just sorry I just wanted to mention that uh The Continuous time counter counter part 4 that's step-by-step paper might be this one from uh 2017 by Rafael bogach because in this paper uh it almost exclusively deals with the continuous time active in prints great generalized coordinates motion this is again bringing back this idea of the Taylor series if this block line is the function that has to be approximated if you only have one parameter to approximate it and you want to do it at this line your first is just the actual value of the function at T equals zero τ equals zero then the first derivative value at the time x of 0 the adding in the second derivative and so on so as you add more and more and more derivatives you'll get better and better approximation to arbitrary functions and you'll get that with a totally known form that can be represented in a vector as the generalized coordinates of motion and in physics it's it's um used in engineering it's used in PID control this has come up in many live streams and discussions because you can say whatever function I'm approximating I want the first five derivatives and then you know that your data structure is only going to have five coefficients for example and they've joked many times like six generalized coordinates are sufficient six layers of the brain cortex regions though in some ways that's kind of that makes it seem like the six layers of the cortex are like this but actually the cortical column is not like little six floors where each one is its own predictive Loop actually the layers are connected differently formalisms in the continuous time setting with generalized coordinates of motion $\tilde{x}$ over a variable is representing trajectories one footnote is that this is still in the um date based base of Bayesian mechanics Ollie or anyone else please correct if this is not the case so the tilde here a summary notation helping us represent trajectories of states which is moving us towards the path formulation of Bayesian mechanics but is not the same as a true path formulation hope that is fair to say the LaPlace approximation is this yes just help me please yeah so in that in the till the formalism now when we are talking about generalized coordinates if we were talking about standard coordinates uh $\tilde{x}$ would be x still there would be x at time T comma x at time t plus 1 x at time t plus two Etc now in the generalized coordinates the $\tilde{x}$ would be x comma first derivative of x comma second derivative of x is that correct yes so oh tilde with just a location coordinate would be either a list or a vector depending on whether you want to you know think about it in terms of like a list with comma or in terms of like a vector going down that time yeah yeah then with generalized coordinates it is a matrix so just it it could be represented differently but here let's just say that we're going to be working with it as a list like you said so in this case columns correspond to the list above those are the time points rows

going down are the generalized coordinates of motions so for each time point one two three four five we have a vector which is describing the location the first derivative the second derivative and so on I see I see so there would be six rows mm-hmm cool if we had six generalized coordinates of motion and um this is used and discussed in um live stream 26 particularly and several other places are like they're kind of a uh a little not not like a a trick or anything to get Dynamics into a static representation like if you just take a picture of a car you can't tell if it's moving or not but if you said this is a generalized picture of a car it's at um it's at 12 on the number line and its velocity is 10 this way and its acceleration it's slowing down and it's going to the right and the way that it's slowing down is speeding up then you can start to say a lot more about the next steps of the car exactly and that is representable in a static um snapshot Ollie uh and another way to looking at uh the generalized coordinates is to see it as a a dimensional reduction technique because uh when we want to describe as a very simple case a trajectory in a Cartesian coordinates uh well uh we'll definitely need at least um for example in a two-dimensional trajectory we'll need two Dimensions but if we look at it from a generalized coordinates perspective we can just put it into the Riemann manifold and jettison one of the dimensions which encapsulates that trajectory through a specific time so one of the dimensions would be reduced and then we can just look at it trajectory directly look at the behavior of the trajectory directly from the consecutive derivatives we take so that's a dimension reduction technique as well nice point yeah when you said uh you said on the remaining Romanian ring onion manifold he said can you yeah yeah because in the uh Punk array or Romanian manifold uh we're dealing with uh not exactly the Cartesian coordinates but uh it's just the state space defined as these kind of General so it is already a one dimension reduced coordinate so to speak so that's the whole point behind uh this I mean encompassing the state space in that Romanian manifold not in the Cartesian coordinates because Cartesian coordinates need some global coordinates to define the states of the space but in generalized coordinates it's more like a kind of localized uh coordinates not Global ones boom yes this is a very well I don't know I don't know much about the Romanian or or the point manifold so yes I have some readings to to do to understand that but yeah this is very deep and Powerful um analytical Insight one other way to think about generalized coordinates as a dimensionality reduction technique from a more computational applied perspective if you want five coordinates of motion you know exactly the size of that data object if you say I'm going to consider 20 time points and five generalized coordinates of motion you know exactly the shape of that parameter or that variable now whether that captures whether that manifold or

Subspace captures 10 or 90 or 100 percent of the variance in the original space is situational but um you will capture some variants just like every principal component analysis will capture some variance in the space and whether it captures enough is situational you know you get some and you know that you've reduced the dimensionality to a well-defined amount which in many cases is essential versus having like an unbounded number of of unbounded size um the motor reflex case has been especially studied from The Continuous time perspective that's going to come up in the second half of the book it comes up in a lot of different other cases and in chapter motor reflexes box fortree describes the LaPlace approximation oh there's no again as frequently that last names don't really reduce uncertainty about what something is talking about much of the time and so there'd be ways to describe some of these things that wouldn't rely as much on specific last names because there's like a many-to-many mapping between the technique and different last names and so on and um but just working with what we have LaPlace approximation is a quadratic approximator that is an expansion around the mode of a distribution so the procedure for generating a laplacian approximation is you find the mode of a distribution which is the tallest point on the distribution an upside down Parabola is like fit to that mode the tip of the parabola is going to be the the mode of the distribution the width of the parabola is going to reflect a second parameter a variance parameter that captures as much as possible of the original distribution so this is kind of like created if um a good thing about LaPlace approximation is that no matter what shape the original distribution has you know your LaPlace approximation is only going to need two parameters the mode and the variants and so in that way it's a lot like a gaussian approximator although the gaussian function is different than the quadratic approximator um they're similar because they both are um have two parameters the location of the midpoint and how wide the function should be stretched for empirical distributions which might not fit cleanly to any specific analytical family this can be very helpful because again you know that some density of the target distribution will be captured now if you're dealing with like a super tricky distribution where there's four humps all of very similar height then the LaPlace approximation is going to get tricked because it's only going to capture like at best 25 percent of the variance and it's gonna you're gonna be very much um like unaware of these other three big Peaks however four distributions that have a central tendency broadly speaking the LaPlace approximation does extremely well some of these subtleties between like LaPlace approximation variational Bays and non-parametric bootstrap based sampling techniques are explored in the SPM textbook especially all of this was still inferential but not action including

and that is very analogous to the approach taken in the paper discussed in uh live stream 43 which is on predictive processing itself predictive coding a theoretical experimental review in this one eighty percent or something in the paper is about hierarchical predictive processing architectures and then they go all right now we're just going to slip in action and it turns out to play very well here we see a figure that's kind of resonant with this one we have a discrete time formulation here we know that because we see like an explicit time variable and like a time minus one and time plus one so that's how one knows that they're dealing with a discrete time formulation even if the variables were like called different things um in contrast we see a hierarchical architecture again everything has been transposed into the notation of the continuous time where we saw like why coming in whereas we saw o coming in here and that was what we saw in four three o coming in to discrete time y coming into continuous so this figure four six is like related to the previous message passing figure and then there's a summary chapter four as Ali mentioned at the beginning relatively technical there's a lot of techniques and notation and forays into more General topics ranging from Jensen's inequality at the beginning of the chapter to generalize coordinates of motion message passing in the context of Markov blankets evidence lower bound and free energy calculations and all of that is being done in this side-by-side way between discrete and continuous time and that is um broadly what chapter four describes all of that is being done because we're exploring generative models in active inference okay in our final minutes what are some thoughts or ideas that people have that we can come to next week uh Yvonne I'm past now what does it mean to understand chapter four what are we looking to get out of chapter four the discrete and continuous time distinction is like in my opinion the essence of of what is in chapter four um but I would like to like understand more about the message passing because I don't really get it all the way as I would also love to like pick apart and understand the the larger paper on message passing by Kristen and some colleagues that I don't remember but for me the greatest challenge is that you know even though you you're beginning to to grasp the theoretical uh details the the moving from there to actually implementing something is is still uh still quite difficult um and uh you know Freestone says a quartz a lot of time fine man saying you can only understand when you start when you can do it uh but still it's uh that passage in active inference is uh is still quite uh challenging for me I mean even after after having read the whole book and you know and a few of the other papers uh so this is the even though you know you you understand this this for chapter four which is a lot of the theory but still going from there to implementing a model it's a not easy I don't know if there's something that uh

affects everybody it does affect me for sure certainly it does it's the fractal state of play at this moment that's why people aren't just like blasting out active models because there's like a lot of uh trails to blaze and also just as always compare it with a linear regression you could read a 100 500 10 000 page textbook on linear and then you would sit down in front of your programming language and you may not specifically know how to quote apply linear regression linear regression has had a hundred years of empirically focused development on data input output visualization training model adequacy model selection there are just whole areas of Applied statistics women haven't turned their attention to this area which is why we're doing it so the most ancient comparator is linear regression then to get one step closer to seeing what the theory practice synthesis looks like the SPM textbook documentation and literature help us see what it could look like and then it'll have aspects of both of those and some novelties all right very well thank you all for this fun meeting thank you see you next week thank you bye-bye